

Shraddha Gourishetty

GIS Analyst | Geospatial Data Science | Remote Sensing | Python Automation

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TECHNICAL SKILLS

Esri Platform: ArcGIS Pro, ArcGIS Online, ArcGIS Enterprise, ArcMap, Portal, Dashboards, Experience Builder

Python: ArcPy, Pandas, NumPy, Scikit-Learn, GeoPandas, Rasterio, Matplotlib, PyTorch, GEE API

Databases: PostgreSQL/PostGIS, SQL, ArcSDE geodatabases, spatial data modelling, ETL workflows

Machine Learning: Random Forest, supervised/unsupervised classification, feature engineering, cross-validation

Remote Sensing: Google Earth Engine, satellite imagery, land cover classification, LiDAR, raster analysis, DEM/DTM

Photogrammetry: DJI Terra, GCP processing, orthomosaic, DSM/DTM (Transport Canada Drone Cert)

Web GIS: StoryMaps, Instant Apps, web map development, JavaScript, HTML

Field Tools: Field Maps, Survey123, QuickCapture, GNSS data collection

PROFESSIONAL EXPERIENCE

GIS Business Analyst | Korotu Technology | Toronto, ON

Jan 2024 – Aug 2024

- Ran end-to-end drone photogrammetry workflows (GCP processing, orthomosaic, DSM/DTM) and integrated outputs into ArcGIS/QGIS pipelines for climate and land-use analysis
- Built Python-automated data pipelines consolidating 5+ environmental datasets, reducing analysis time 30%; produced 15+ map outputs supporting \$150K–\$200K in awarded funding

GIS Analyst | Region of Peel | Mississauga, ON

May 2023 – Aug 2023

- Spatial analysis on 10+ years of surveillance data for 1.5M residents; built predictive models using GIS and R integrating environmental, demographic, and land-use datasets

Data and Analytics Technician | Regional Municipality of York | Newmarket, ON

Sep 2022 – Dec 2022

- Built a custom ArcPy toolbox automating enterprise geodatabase QA/QC, reducing processing time ~40% across nine municipalities in a live ArcGIS Enterprise environment
- Published and maintained 10+ spatial services through ArcGIS Enterprise Portal and ArcGIS Online; developed an interactive JavaScript web map for non-technical municipal staff

TECHNICAL PROJECTS

Walkable Toronto: LiDAR-Based Accessibility and Urban Heat Risk Platform | *In Progress*

- Using LiDAR point cloud data to map Toronto streets by walkability difficulty and urban heat vulnerability, combining tree canopy, sidewalk conditions, and heat island analysis

Downtown Toronto Traffic Intelligence and Risk Prediction | *Python, ArcGIS Pro, ML*

- Built spatial risk models using Toronto Open Data and 500m grid aggregation; applied ML clustering to identify high-risk traffic zones; delivered via interactive ArcGIS StoryMap

Wildfire Risk Classification Model | *Scikit-Learn, ArcGIS Pro, ETL*

- Random Forest classifier at 96% accuracy integrating terrain, vegetation, and weather raster datasets with feature engineering and cross-validation; tiered risk maps for emergency planning stakeholders

Land Cover and Soil Moisture Analysis | *Google Earth Engine, Python*

- Processed 100+ satellite scenes through automated GEE Python pipelines; Standardized Soil Moisture Index applied to assess desertification risk across Sudan-Sahel

Fire Station Network Analysis and Response Optimization | *ArcGIS Pro, Network Analysis*

- Built a road network dataset with full attribution; generated service area and location-allocation models; produced spatial decision support outputs for Waterloo Region planners

EDUCATION

Bachelor of Environmental Studies, Geomatics (Honours) | University of Waterloo | Waterloo, ON

Sep 2020 – Jun 2025

Specialization in Aviation | Minor in Computing | Coursework: GIS Programming, Spatial Databases, Remote Sensing, LiDAR

CERTIFICATIONS

Transport Canada Basic Drone Pilot Certificate | AutoCAD Map 3D Essential Training (2025) | Google Data Analytics Professional Certificate | ESRI MOOC: Modern Geo Apps